

LOW STYRENE SBR PATENT LANDSCAPE: SOLUTION AND EMULSION ROUTES, FUNCTIONALIZATION, AND TIRE-COMPOUND PERFORMANCE

1. ABSTRACT

This report synthesizes six primary patents relevant to styrene-butadiene rubber (SBR) and styrene-butadiene copolymer synthesis, with emphasis on embodiments that can support low-styrene grades or composition control. The selected patents span both solution polymerization and emulsion polymerization, plus downstream functionalization strategies that improve tire-relevant properties without changing the core SBR identity. Across the landscape, the dominant synthesis themes are anionic solution polymerization in hydrocarbon solvent with *n*-butyllithium initiation, controlled monomer feed of styrene and 1,3-butadiene, terminal modification or coupling, and post-polymerization functionalization using silane-, carbinol-, or carboxyl-containing reagents. In parallel, emulsion-state radical polymerization in water is used to prepare functional SBRs and related copolymers, often with emulsifiers, redox initiators, mercaptan chain-transfer agents, and aqueous stripping/drying workup.

The target definition here is low styrene fraction in SBR, but the extracted evidence rarely states a dedicated low-styrene claim; instead, it provides explicit styrene contents or feed ratios for specific SBR embodiments. The most compositionally informative disclosures are EP3056520B1 and EP2989129B1, which report styrene contents around 24.7-25.4 wt% for solution-polymerized, silane-containing or carbinol-terminated SBRs, and EP3070120B1 and US9328176B2, which disclose emulsion SBR systems with styrene feeds or resulting styrene contents ranging from 20 wt% to 40 wt% depending on embodiment. US10174133B2 and US5902856A focus more on solution blending, terminal modification, and molecular-weight architecture than on styrene fraction itself, but they remain relevant because they disclose SBR synthesis and downstream processing routes used in tire compounds.

Overall, the patent set shows two major process trends: solution SBR routes that enable tighter control of microstructure, molecular weight, and terminal functionality, and emulsion SBR routes that use water as the polymerization medium and are readily adapted to polar-functional monomers. Formulation trends emphasize silica-filled tread compounds, use of silanes and coupling agents, and property optimization for rolling resistance, wet grip, abrasion resistance, and

processability.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US10174133B2
- Patent Title: Modified styrene-butadiene copolymer, preparation method thereof, and rubber ...
- Assignee: Lg Chem, Ltd.
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses solution polymerization of styrene and 1,3-butadiene in normal hexane using n-butyllithium in an autoclave reactor, followed by terminal capping with additional butadiene and post-polymerization modification with a silane-containing modifier. The polymer is then terminated with ethanol, stabilized with antioxidant, and isolated by warm-water solvent removal and roll drying. - Styrene: 29.7 g — Example 1 - 1,3-Butadiene: 78.1 g — Example 1 - Normal hexane: 550 g — Example 1 - 2,2-di(2-tetrahydrofuryl)propane (DTP): 0.124 g — Example 1 - Reaction temperature: 50° C. — Example 1 - n-Butyllithium: 3 mmol — Example 1 - Terminal capping butadiene: 2.2 g — Example 1 - Modifier: 0.181 g — Example 1 - Modifier/[act. Li]: 1.0 — Example 1 - [DTP]/[act. Li]: 2.3 — Example 1 - Styrene: 270 g — Example 2 - 1,3-Butadiene: 710 g — Example 2 - Normal hexane: 5 kg — Example 2 - 2,2-di(2-tetrahydrofuryl)propane (DTP): 0.94 g — Example 2 - Reaction temperature: 40° C. — Example 2 - n-Butyllithium: 25.40 g (2.62 wt % in hexane, 33% activation) — Example 2 - Terminal capping butadiene: 20.0 g — Example 2 - Modifier: 1.98 g — Example 2 - Modifier/[act. Li]: 0.85 — Example 2 - [DTP]/[act. Li]: 1.46 — Example 2 - Antioxidant solution: 4.5 ml of 0.3 wt % BHT in hexane — Example 1 - Antioxidant solution: 33 g of 30 wt % Wingstay K in hexane — Example 2 - Medium & Water Role: The core polymerization medium is hydrocarbon solution polymerization in normal hexane. Water is used later only for warm-water solvent removal and downstream drying/roll drying, not as the polymerization medium. - low styrene fraction: Not disclosed in extracted evidence

Medium & Water Role - The core polymerization medium is hydrocarbon solution polymerization in normal hexane. Water is used later only for warm-water solvent removal and downstream drying/roll drying, not as the polymerization medium. - Core reaction medium: Organic hydrocarbon solution polymerization in normal hexane

low styrene fraction - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1: Prepared a modified styrene-butadiene copolymer in hexane at 50° C. with n-butyllithium, terminal butadiene capping, and silane-modifier incorporation. - Example 2: Demonstrated the same solution-modification route at larger scale and 40° C., using a different silane modifier and higher modifier loading. - Comparative Example: Showed that changing the modifier to N,N-bis(triethoxysilylpropyl)piperazine still gave a modified conjugated diene copolymer by the same base route. - Experimental Example 1: Compared molecular-weight and combination-efficiency data for the modified copolymers from Examples 1, 2, and Comparative Example.

Technical Relevance This patent is relevant because it is a direct SBR synthesis and modification disclosure based on solution polymerization of styrene and butadiene. Although it does not target low styrene explicitly, it provides a controllable SBR platform and explicit monomer feed quantities that define the copolymer composition.

Patent 2

- Patent Number: EP3070120B1
- Patent Title: Polymer composite and method for producing the same, and rubber composition for ...
- Assignee: Sumitomo Rubber Industries, Ltd.
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses emulsion-state radical polymerization of styrene, butadiene, and an epoxy acrylate monomer in water using emulsifiers, EDTA/ferrous sulfate redox components, n-dodecylmercaptan, and p-menthane hydroperoxide, followed by ring-opening under acidic or alkaline conditions. The resulting functional SBR is then stripped and dried and used in silica-filled tread compositions. - Water: 1,500 mL — Example 1 - Sodium rosinat: 25 g — Example 1 - Sodium salt of fatty acid: 35 g — Example 1 - Styrene: 380 g — Example 1 - Glycidyl methyl methacrylate: 50 g — Example 1 - 1,3-Butadiene: 570 g — Example 1 - EDTA: 0.6 g — Example 1 - Ferrous sulfate: 0.1 g — Example 1 - n-Dodecylmercaptan: 1.5 g — Example 1 - p-Menthane hydroperoxide: 0.6 g — Example 1 - Reaction temperature: 10° C. — Example 1 - Reaction time: 24 hours — Example 1 - Diethylhydroxylamine: 1.0 g — Example 1 - Sulfuric acid: 20% aqueous solution, 100 g — Example 1 - Average molecular weight: 908,000 g/mol — Example 1 - Water: 1,500 mL — Comparative Example 1 - Styrene: 400 g — Comparative Example 1 - 1,3-Butadiene: 600 g — Comparative Example 1 - Reaction temperature: 10° C. — Comparative Example 1 - Reaction time: 24 hours — Comparative Example 1 - Average molecular weight: 843,000 g/mol — Comparative Example 1 - Water: 1,500 mL — Comparative Example 2 - Styrene: 380 g —

Comparative Example 2 - 1,3-Butadiene: 570 g — Comparative Example 2 - Methyl methacrylic acid: 50 g — Comparative Example 2 - Reaction temperature: 10° C. — Comparative Example 2 - Reaction time: 24 hours — Comparative Example 2 - Average molecular weight: 795,000 g/mol — Comparative Example 2 - Styrene content: 40% by mass — Polymer 1 - Styrene content: 20% by mass — Polymer 2 - Styrene content: 40% by mass — Polymer 3 - Styrene content: 45% by mass — Polymer 4 - Styrene content: 50% by mass — Polymer 5 - Medium & Water Role: Water is part of the core polymerization medium in the emulsion-state radical polymerization. Water is also used later in stripping/drying and in the aqueous ring-opening step with sulfuric acid, NaOH, or KOH. - low styrene fraction: 20% by mass

Medium & Water Role - Water is part of the core polymerization medium in the emulsion-state radical polymerization. Water is also used later in stripping/drying and in the aqueous ring-opening step with sulfuric acid, NaOH, or KOH. - Core reaction medium: Aqueous emulsion polymerization

low styrene fraction - 20% by mass

Relevant Experimental Evidence - Example 1: Prepared a styrene-butadiene-dihydroxypropyl methacrylate copolymer in aqueous emulsion and ring-opened it with sulfuric acid. - Example 4: Showed that NaOH can be used as the ring-opening agent to obtain the same functional copolymer family. - Example 5: Showed that KOH can also be used as the ring-opening agent. - Comparative Examples 1 and 2: Prepared non-functional and methyl-methacrylic-acid-containing SBRs under the same emulsion conditions. - Polymer list: Provided explicit styrene contents for several SBR and related copolymers used in tread formulations.

Technical Relevance This patent is highly relevant because it is a direct SBR synthesis disclosure and explicitly uses an aqueous emulsion route, which is one of the main industrial pathways for SBR. It also provides explicit styrene-content values for several SBR embodiments, including a 20% by mass polymer that is compositionally closer to a lower-styrene grade.

Patent 3

- Patent Number: US9328176B2
- Patent Title: Functional styrene-butadiene copolymer
- Assignee: Korea Kumho Petrochemical Co., Ltd.
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses emulsion polymerization of styrene and 1,3-butadiene in water with sodium rosinatate and sodium fatty-acid soap emulsifiers, EDTA/ferrous sulfate redox components, n-dodecylmercaptan chain-transfer agent, and p-menthane hydroperoxide initiator. After polymerization at 10° C. for 24 hours, the

latex is terminated with diethylhydroxylamine, then stripped and dried; some embodiments are further ring-opened with aqueous sulfuric acid, NaOH, or KOH to introduce hydroxyl functionality. - Water: 1,500 mL — Comparative Example 1 - Sodium rosinate: 25 g — Comparative Example 1 - Sodium salt of fatty acid: 35 g — Comparative Example 1 - Styrene: 400 g — Comparative Example 1 - 1,3-Butadiene: 600 g — Comparative Example 1 - EDTA: 0.6 g — Comparative Example 1 - Ferrous sulfate: 0.1 g — Comparative Example 1 - n-Dodecylmercaptan: 1.5 g — Comparative Example 1 - p-Menthane hydroperoxide: 0.6 g — Comparative Example 1 - Reaction temperature: 10° C. — Comparative Example 1 - Reaction time: 24 hours — Comparative Example 1 - Diethylhydroxylamine: 1.0 g — Comparative Example 1 - Average molecular weight: 843,000 g/mol — Comparative Example 1 - Water: 1,500 mL — Comparative Example 2 - Sodium rosinate: 25 g — Comparative Example 2 - Sodium salt of fatty acid: 35 g — Comparative Example 2 - Styrene: 380 g — Comparative Example 2 - 1,3-Butadiene: 570 g — Comparative Example 2 - Methyl methacrylic acid: 50 g — Comparative Example 2 - Reaction temperature: 10° C. — Comparative Example 2 - Reaction time: 24 hours — Comparative Example 2 - Average molecular weight: 795,000 g/mol — Comparative Example 2 - Water: 1,500 mL — Example 1 - Sodium rosinate: 25 g — Example 1 - Sodium salt of fatty acid: 35 g — Example 1 - Styrene: 380 g — Example 1 - Glycidyl methyl methacrylate: 50 g — Example 1 - 1,3-Butadiene: 570 g — Example 1 - EDTA: 0.6 g — Example 1 - Ferrous sulfate: 0.1 g — Example 1 - n-Dodecylmercaptan: 1.5 g — Example 1 - p-Menthane hydroperoxide: 0.6 g — Example 1 - Reaction temperature: 10° C. — Example 1 - Reaction time: 24 hours — Example 1 - Diethylhydroxylamine: 1.0 g — Example 1 - Sulfuric acid: 20% aqueous solution, 100 g — Example 1 - Average molecular weight: 908,000 g/mol — Example 1 - NaOH: 20% aqueous solution, 100 g — Example 4 - Average molecular weight: 740,000 g/mol — Example 4 - KOH: 20% aqueous solution, 100 g — Example 5 - Average molecular weight: 791,000 g/mol — Example 5 - Medium & Water Role: Water is the core polymerization medium in the emulsion process. Water is also used later in the aqueous ring-opening step and in stripping/drying of the latex-derived polymer. - low styrene fraction: Not disclosed in extracted evidence

Medium & Water Role - Water is the core polymerization medium in the emulsion process. Water is also used later in the aqueous ring-opening step and in stripping/drying of the latex-derived polymer. - Core reaction medium: Aqueous emulsion polymerization

low styrene fraction - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1: Prepared a glycidyl methyl methacrylate-functionalized SBR latex and ring-opened it with sulfuric acid to give a hydroxylated copolymer. - Example 4: Showed that NaOH can be used as the ring-opening agent to obtain the hydroxylated copolymer. - Example 5: Showed that KOH can be used as the ring-opening agent to obtain the hydroxylated copolymer. - Comparative Example 1: Prepared an unfunctionalized SBR under the same emulsion conditions. - Comparative Example 2: Prepared a styrene-butadiene-methacrylic acid copolymer under the same emulsion conditions.

Technical Relevance This patent is directly relevant because it discloses a conventional emulsion SBR process with explicit monomer and emulsifier conditions, which is a core manufacturing route

for low-styrene SBR grades. The examples also show how composition can be tuned through styrene and comonomer feed choices in an aqueous latex process.

Patent 4

- Patent Number: US5902856A
- Patent Title: Rubber composition of two diene rubbers
- Assignee: Nippon Zeon Co., Ltd.
- Jurisdiction: US
- Publication Year: 1999
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses solution blending of a terminal-modified polymer-containing solution with a high-molecular polymer-containing solution, followed by steam stripping for coagulation and recovery, water squeezing, and vacuum drying to obtain a diene rubber. The diene rubber is then compounded into rubber compositions and vulcanized for property evaluation. - Stirring temperature: 60° C. — Example 1 - Stirring time: 30 minutes — Example 1 - 2,6-di-t-butyl-p-cresol: 10 g — Example 1 - Press vulcanization temperature: 160° C. — Example 1 - Press vulcanization time: 25 minutes — Example 1 - Material rubber: 100 parts — Example 1 - Stearic acid: 1.5 parts — Example 1 - Zinc oxide #1: 3 parts — Example 1 - N-339: 50 parts — Example 1 - Aromatic oil: 5 parts — Example 1 - Sulfur: 1.75 parts — Example 1 - Vulcanization accelerator: 1 part — Example 1 - Terminal-modified polymer A: ST/V 21/63, M1 50000 — Table 1 - Terminal-modified polymer B: ST/V 21/66, M1 150000 — Table 1 - Terminal-modified polymer C: ST/V 21/65, M1 210000 — Table 1 - Terminal-modified polymer D: ST/V 21/65, M1 250000 — Table 1 - Terminal-modified polymer E: ST/V 21/65, M1 300000 — Table 1 - Terminal-modified polymer F: ST/V 21/65, M1 360000 — Table 1 - Terminal-modified polymer G: ST/V 21/64, M1 650000 — Table 1 - High-molecular polymer H: ST/V 21/65, M2 1200000 — Table 1 - High-molecular polymer I: ST/V 22/66, M2 900000 — Table 1 - High-molecular polymer J: ST/V 21/64, M2 600000 — Table 1 - Molecular weight ratio (M2/M1): 5.5 — Table 2 - Molecular weight ratio (M2/M1): 4.8 — Table 3 - Molecular weight ratio (M2/M1): 4.8 — Table 5 - Molecular weight ratio (M2/M1): 2.9 — Table 7 - Medium & Water Role: The core process is solution-state blending of polymer solutions, not aqueous polymerization. Water appears later in steam stripping for coagulation and in water squeezing during recovery of the rubber. - low styrene fraction: Not disclosed in extracted evidence

Medium & Water Role - The core process is solution-state blending of polymer solutions, not aqueous polymerization. Water appears later in steam stripping for coagulation and in water squeezing during recovery of the rubber. - Core reaction medium: Organic solution blending / solution-state polymer processing

low styrene fraction - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1: Showed that mixing terminal-modified and high-molecular polymer solutions in solution state gives a diene rubber recovered by steam stripping and water squeezing. - Example 2: Demonstrated that changing the modifier in the production examples still yields similar diene rubber performance. - Example 3: Showed that different styrene contents and 1,2-bond contents can be combined while retaining improved processability and abrasion resistance. - Example 5: Showed that silica-filled tread compounds made from the mixed diene rubber exhibit improved abrasion resistance and resiliency. - Example 6: Showed that the mixed diene rubber can be blended with natural rubber, emulsion SBR, and/or high-cis polybutadiene.

Technical Relevance This patent is relevant because it discloses solution-state SBR-containing diene rubber architectures and explicitly reports styrene-to-vinyl ratios for several polymers. While not a low-styrene patent per se, it provides compositionally defined SBR-related materials and downstream processing routes used in tire compounds.

Patent 5

- Patent Number: EP2989129B1
- Patent Title: Carboxyl-terminated polymers containing silanes
- Assignee: ARLANXEO Deutschland GmbH
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses anionic solution polymerization of styrene-butadiene copolymers followed by end-group functionalization with silane-containing carboxyl groups. The examples use hexane, styrene, 1,3-butadiene, 2,2-bis(2-tetrahydrofuryl)propane, and butyllithium, then terminate with steam stripping and vacuum drying. - Hexane: 8.5 kg — Example 1 - 1,3-Butadiene: 1185 g — Example 1 - Styrene: 315 g — Example 1 - 2,2-Bis(2-tetrahydrofuryl)propane: 8.6 mmol — Example 1 - Butyllithium: 11.3 mmol — Example 1 - Reaction temperature: 60° C. — Example 1 - Reaction time: 25 minutes — Example 1 - Cetyl alcohol: 11.3 mmol — Example 1 - Irganox 1520: 3 g — Example 1 - Steam stripping: used — Example 1 - Vacuum drying temperature: 65° C. — Example 1 - Hexamethylcyclotrisiloxane: equimolar to butyllithium — Example 2 - Additional heating: 20 minutes at 60° C. — Example 2 - 2,2-Dimethyl-1-oxa-4-thia-2-silacyclohexan-6-one: equimolar to butyllithium and hexamethylcyclotrisiloxane — Example 3 - Additional heating: 20 minutes at 60° C. — Example 3 - Pyrrolidine: equimolar to butyllithium — Example 4 - 2,2,4-Trimethyl-1-oxa-4-aza-2-silacyclohexane: equimolar to butyllithium — Example 5 - Styrene content: 24.7% — Example 1a - Vinyl content: 51.5% — Example 1a - Tg: -16°C — Example 1a - Mn: 242 kg/mol — Example 1a - Mw/Mn: 1.30 — Example 1a - Mooney viscosity: 71 ME — Example 1a - Styrene content: 24.8% — Example 1b - Vinyl content: 51.2% — Example 1b - Tg: -15°C — Example 1b - Mn: 261 kg/mol — Example 1b - Mw/Mn: 1.17 — Example 1b - Mooney

viscosity: 73 ME — Example 1b - Styrene content: 25.4% — Example 1c - Vinyl content: 51.3% — Example 1c - Tg: -17°C — Example 1c - Mn: 233 kg/mol — Example 1c - Mw/Mn: 1.18 — Example 1c - Mooney viscosity: 69 ME — Example 1c - Medium & Water Role: The core polymerization medium is hexane solution polymerization. Water is used later only in steam stripping for solvent removal and in vacuum-drying workup, not as the polymerization medium. - low styrene fraction: 24.7%

Medium & Water Role - The core polymerization medium is hexane solution polymerization. Water is used later only in steam stripping for solvent removal and in vacuum-drying workup, not as the polymerization medium. - Core reaction medium: Organic hydrocarbon solution polymerization in hexane

low styrene fraction - 24.7%

Relevant Experimental Evidence - Example 1a: Prepared an unfunctionalized styrene-butadiene copolymer in hexane at 60° C. and isolated it by steam stripping and vacuum drying. - Example 1b: Added 2,2-dimethyl-1-oxa-2-silacyclohexane after polymerization to obtain a silane-containing carbinol-terminated SBR. - Example 1c: Added 2,2,4-trimethyl-1-oxa-4-aza-2-silacyclohexane to obtain another silane-containing carbinol-terminated SBR. - Examples 2a-c: Formulated tread compounds containing the SBRs and evaluated vulcanizate properties. - Examples 3a-c: Showed improved rebound resilience, tan delta, and abrasion for the inventive silane-containing SBRs.

Technical Relevance This patent is directly relevant because it discloses solution-polymerized SBR with explicit styrene contents around 25 wt%, which is within a low-to-moderate styrene regime for tire SBR. It also shows how end-group functionalization can be layered onto a compositionally defined SBR to improve tread performance.

Patent 6

- Patent Number: EP3056520B1
- Patent Title: Silane-containing carbinol-terminated polymers
- Assignee: ARLANXEO Deutschland GmbH
- Jurisdiction: EP
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses anionic solution polymerization of styrene and 1,3-butadiene in hexane using butyllithium and 2,2-bis(2-tetrahydrofuryl)propane, followed by end-group functionalization with silane-containing carbinol-forming reagents. The polymer solution is then stabilized, steam-stripped, and vacuum-dried to isolate the SBR. - Hexane: 8.5 kg — Example 1a - 1,3-Butadiene: 1125 g — Example 1a - Styrene: 375 g — Example 1a - 2,2-Bis(2-tetrahydrofuryl)propane: 28 mmol —

Example 1a - Butyllithium: 10 mmol — Example 1a - Reaction temperature: 70° C. — Example 1a - Reaction time: 1 h — Example 1a - Irganox 1520: 3 g — Example 1a - Vacuum drying temperature: 65° C. — Example 1a - 2,2-Dimethyl-1-oxa-2-silacyclohexane: 11 mmol (1.43 g) — Example 1b - Additional heating: 20 min at 70° C. — Example 1b - 2,2,4-Trimethyl-1-oxa-4-aza-2-silacyclohexane: 11.6 mmol (1.69 g) — Example 1c - Additional heating: 20 min at 70° C. — Example 1c - Styrene content: 24.7% — Example 1a - Vinyl content: 51.5% — Example 1a - Tg: -16°C — Example 1a - Mn: 242 kg/mol — Example 1a - Mw/Mn: 1.30 — Example 1a - Mooney viscosity: 71 ME — Example 1a - Styrene content: 24.8% — Example 1b - Vinyl content: 51.2% — Example 1b - Tg: -15°C — Example 1b - Mn: 261 kg/mol — Example 1b - Mw/Mn: 1.17 — Example 1b - Mooney viscosity: 73 ME — Example 1b - Styrene content: 25.4% — Example 1c - Vinyl content: 51.3% — Example 1c - Tg: -17°C — Example 1c - Mn: 233 kg/mol — Example 1c - Mw/Mn: 1.18 — Example 1c - Mooney viscosity: 69 ME — Example 1c - Medium & Water Role: The core polymerization medium is hexane solution polymerization. Water is used later only in steam stripping for solvent removal and in vacuum-drying workup, not as the polymerization medium. - low styrene fraction: 24.7%

Medium & Water Role - The core polymerization medium is hexane solution polymerization. Water is used later only in steam stripping for solvent removal and in vacuum-drying workup, not as the polymerization medium. - Core reaction medium: Organic hydrocarbon solution polymerization in hexane

low styrene fraction - 24.7%

Relevant Experimental Evidence - Example 1a: Prepared a baseline styrene-butadiene copolymer in hexane at 70° C. with butyllithium initiation. - Example 1b: Introduced 2,2-dimethyl-1-oxa-2-silacyclohexane to obtain a silane-containing carbinol-terminated SBR. - Example 1c: Introduced 2,2,4-trimethyl-1-oxa-4-aza-2-silacyclohexane to obtain another silane-containing carbinol-terminated SBR. - Examples 2a-c: Formulated tread compounds using the SBRs and evaluated compound properties. - Examples 3a-c: Showed improved rebound resilience, tan delta, and abrasion for the inventive SBRs.

Technical Relevance This patent is directly relevant because it discloses solution-polymerized SBR with explicit styrene content around 25 wt%, which is a compositionally useful low-styrene regime. The patent also demonstrates how end-group silane functionality can be added while preserving the SBR backbone and tire-tread utility.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	low styrene fraction	Key Findings
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US10174133B2	Lg Chem, Ltd.	2019	The core polymerization medium is hydrocarbon solution polymerization in normal hexane. Water is used later only for warm-water solvent removal and downstream drying/roll drying, not as the polymerization medium.	Not disclosed in extracted evidence	Example 1: Prepared a modified styrene-butadiene copolymer in hexane at 50° C. with n-butyllithium, terminal butadiene c
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EP3070120B1	Sumitomo Rubber Industries, Ltd.	2019	Water is part of the core polymerization medium in the emulsion-state radical polymerization. Water is also used later in stripping/drying and in the aqueous ring-opening step with sulfuric acid, NaOH, or KOH.	20% by mass	Example 1: Prepared a styrene-butadiene-dihydro methacrylate copolymer in aqueous emulsion and ring-opened it with
US9328176B2	Korea Kumho Petrochemical Co., Ltd.	2016	Water is the core polymerization medium in the emulsion process. Water is also used later in the aqueous ring-opening step and in stripping/drying of the latex-derived polymer.	Not disclosed in extracted evidence	Example 1: Prepared a glycidyl methyl methacrylate-functionalized SBR latex and ring-opened it with sulfuric acid to give

US5902856A	Nippon Zeon Co., Ltd.	1999	The core process is solution-state blending of polymer solutions, not aqueous polymerization. Water appears later in steam stripping for coagulation and in water squeezing during recovery of the rubber.	Not disclosed in extracted evidence	Example 1: Showed that mixing terminal-modified and high-molecular polymer solutions in solution state gives a diene rub
EP2989129B1	ARLANXEO Deutschland GmbH	2017	The core polymerization medium is hexane solution polymerization. Water is used later only in steam stripping for solvent removal and in vacuum-drying workup, not as the polymerization medium.	24.7%	Example 1a: Prepared an unfunctionalized styrene-butadiene copolymer in hexane at 60° C. and isolated it by steam stripp

EP3056520B1	ARLANXEO Deutschland GmbH	2020	The core polymerization medium is hexane solution polymerization. Water is used later only in steam stripping for solvent removal and in vacuum-drying workup, not as the polymerization medium.	24.7%	Example 1a: Prepared a baseline styrene-butadiene copolymer in hexane at 70° C. with butyllithium initiation.
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3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Solution-polymerized SBR patents US10174133B2, EP2989129B1, and EP3056520B1 use hydrocarbon solvents (hexane/normal hexane) and n-butyllithium-based anionic initiation, whereas EP3070120B1 and US9328176B2 use aqueous emulsion polymerization with water, emulsifiers, and redox/organic peroxide initiators.
- Explicit styrene-content disclosures cluster around about 24.7-25.4 wt% in EP2989129B1 and EP3056520B1, while EP3070120B1 discloses a lower 20 wt% styrene SBR embodiment (Polymer 2) and higher 40-50 wt% polymers for other embodiments.
- US9328176B2 and EP3070120B1 both use 1,500 mL water and 24 h at 10° C. in emulsion examples, but US9328176B2 emphasizes functionalization via epoxy-acrylate ring opening, whereas EP3070120B1 emphasizes silica-compatible functional SBR with glycidyl methacrylate.
- US10174133B2 focuses on modifier chemistry and terminal butadiene capping rather than styrene fraction control, while US5902856A focuses on blending terminal-modified and high-molecular polymers and reports ST/V ratios rather than a direct low-styrene target.
- Water is central to the polymerization medium in EP3070120B1 and US9328176B2, but in US10174133B2, EP2989129B1, EP3056520B1, and US5902856A it appears only in downstream stripping, coagulation, or drying steps.

4. CONCLUSION

Across the selected primary patents, low-styrene SBR is most directly supported by the solution-polymerized examples in EP2989129B1 and EP3056520B1, which explicitly disclose styrene contents around 24.7-25.4 wt%, and by the emulsion route in EP3070120B1, which includes a 20 wt% styrene SBR embodiment. The remaining patents are relevant because they establish industrially important SBR synthesis and processing platforms—solution anionic polymerization, emulsion polymerization, terminal modification, and silica-compound formulation—that can be used to tune composition and performance. Overall, the evidence shows that low-styrene SBR is achieved through controlled monomer feed and process selection rather than through a single dedicated low-styrene claim, with tire-performance optimization driven by functionalization, molecular-weight control, and downstream compounding.

5. REFERENCES

- US10174133B2 | Modified styrene-butadiene copolymer, preparation method thereof, and rubber ... | Lg Chem, Ltd. | US | 2019 | <https://patents.google.com/patent/US10174133B2/en>
- EP3070120B1 | Polymer composite and method for producing the same, and rubber composition for ... | Sumitomo Rubber Industries, Ltd. | EP | 2019 | <https://patents.google.com/patent/EP3070120B1/en>
- US9328176B2 | Functional styrene-butadiene copolymer | Korea Kumho Petrochemical Co., Ltd. | US | 2016 | <https://patents.google.com/patent/US9328176B2/en>
- US5902856A | Rubber composition of two diene rubbers | Nippon Zeon Co., Ltd. | US | 1999 | <https://patents.google.com/patent/US5902856A/en>
- EP2989129B1 | Carboxyl-terminated polymers containing silanes | ARLANXEO Deutschland GmbH | EP | 2017 | <https://patents.google.com/patent/EP2989129B1/en>
- EP3056520B1 | Silane-containing carbinol-terminated polymers | ARLANXEO Deutschland GmbH | EP | 2020 | <https://patents.google.com/patent/EP3056520B1/en>